

CASE STUDY: EventHub

EventHub is a platform that allows individuals and companies to organize, promote, and manage public and private events.

When a person or a company decides to host events through the platform, they first register as an organizer by providing their name and, if applicable, the name of the company they represent.

The platform assigns each organizer a verified status after completing an internal review process. Alongside their account, every organizer maintains a public-facing profile page where they can write a short biography, link to their personal or company website, and upload a logo.

This profile page is directly tied to the organizer's account and cannot exist without it.

Once their account is set up, an organizer can publish events on the platform. Every event carries a title, a detailed description, a start date, an optional end date, and a maximum number of attendees allowed.

Large-scale events such as annual conferences may also contain smaller, more focused sessions within them. For example, a technology conference might host several breakout workshops on the same day, with each workshop treated as a distinct event that is nested under the main conference event.

Sessions can only belong to one parent event, and a parent event can have any number of sessions.

People who wish to attend can register for any event as attendees. Each attendee provides their full name, an email address, and a home address that includes a street, city, country, and postal code, used for correspondence purposes. Once an attendee has registered for at least one event, the platform issues them a badge — a credential uniquely numbered per attendee, stamped with the date it was issued, and assigned a tier based on their registration history: either Standard or VIP.

A badge belongs to one attendee, and each attendee can have at most one badge.

An attendee can register for multiple events, and any given event will naturally have many attendees signed up for it.

When registering, an attendee may optionally leave a short note to the organizer, and the platform automatically records the exact date and time the registration was completed.

Finally, the system silently tracks when each event record was first created in the database and when it was last modified — these timestamps are maintained by the platform internally and are never displayed on the public event page, nor are they part of the event's data model as visible fields.

Task:

Implement a full EF Core console app applying every configuration technique covered in this session with all entity classes, a DbContext, and OnModelCreating fully configured.

Each entity must be configured using the technique assigned in the session (Data Annotations, Fluent API, or a separate configuration class) — do not mix techniques on the same entity.

ROUTE